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Ahmed Hassan Mostafa

Full Stack Developer (React) & AI Automation

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SKILLS

Languages

TypeScript, JavaScript, Python, Dart, C#, Java (Android bridges)

Frontend

React 18/19, Tailwind CSS, Vite, WebGL2

Backend

Node.js, Express, FastAPI, REST & WebSocket APIs, embedded HTTP servers

Databases

SQLite, IndexedDB, Firebase Firestore

AI & Automation

Gemini API, faster-whisper, NLLB-200, Piper TTS, Silero VAD, CUDA, chatbots, WhatsApp/Telegram/Meta webhooks

Desktop & Mobile

Electron, Flutter (Windows/Android), Capacitor (Android/iOS)

Security & Other

AES-256, TOTP 2FA, Ed25519 licence signing, GIS & georeferencing (proj4), DWG/DXF, installers (Inno Setup, NSIS), automated testing, Git

EDUCATION

Pharos University in Alexandria

B.Sc. Computer Engineering — 3rd Year, double-major track

In progress

LANGUAGES

Arabic Native

English Strong reading & writing, developing speaking

PROFILE

Third-year Computer Engineering student, entirely self-taught, who ships complete products: full-stack web apps with React and Node.js, AI-driven automation, and offline-first desktop and mobile software. 9 projects shipped, including a WebGL2 CAD engine for fibre-network engineers and a real-time voice translator that runs 100% on-device.

PROJECTS

FiberLens — Smart CAD & FTTH Field Engineering Platform

- Offline-first CAD platform that opens and edits DWG/DXF on Android, iOS and the web, with the engineer's live GPS position shown inside the drawing.
- Wrote a WebGL2 renderer that draws 85k entities (~550k segments) in under 1 ms of GPU time; added fibre tracing to the OLT, splitters, georeferencing, BOQ and 10 reports. 27 unit tests.

TypeScript

React

WebGL2

Capacitor

proj4

github.com/Ahshika/FiberLens Live: fiberlens.pages.dev

AI Live Translator — Real-time Two-way Voice Translation

- Fully local pipeline (VAD → Whisper → NLLB-200 → Piper) that translates both sides of a Zoom, Teams or Meet call through a virtual mic, in 103 languages.
- C# helper on the Windows process-loopback API captures only the meeting's audio; CUDA with CPU fallback, one-click installer, 85 tests.

Python

Whisper

NLLB-200

FastAPI

C#

github.com/Ahshika/AI-Live-Translator

FixTrack — Phone Repair Shop & POS System

- One Flutter codebase for Windows and Android: repair workflow (intake → approval → warranty delivery) plus barcode POS, IMEI tracking, used-phone buying and mobile wallets.
- The shop PC runs an embedded HTTP server and SQLite that phones join over Wi-Fi, offline; Firebase QR tracking, Ed25519-signed licences, 72 tests.

Flutter Dart Riverpod SQLite Firebase

github.com/Ahshika/FixTrack

NightGold V3 — Password Manager & File Locker

- Offline Windows password manager unlocked by a TOTP code instead of a master password: AES-256 vault, Arabic-capable generator, weak/reused-password audit.
- Quick Lock encrypts any file or folder in place (scrypt + AES-256) and re-locks it on Windows lock, sleep or shutdown.

Electron React 19 SQLite AES-256 TOTP

github.com/Ahshika/NightGold-Password-Manager

GeoSnap Pro — GPS Camera & Inspection Reports

- Android inspection camera: live stamps in four templates, real EXIF GPS on every photo, drag-the-pin GPS correction.
- In-app gallery, PDF inspection reports, a native Java bridge, signed and obfuscated releases.

Capacitor JavaScript Java Leaflet jsPDF

github.com/Ahshika/GeoSnap-Pro-V2

Password Generator V2 — NightGold V2

- Generator with a live strength meter and a vault (search, favourites, categories) unlocked by TOTP paired through a QR code.

Electron React 19 Tailwind SQLite

github.com/Ahshika/Password-Generator-V2

Password Generator V1 — NightGold V1

- First version: 8–64 character passwords (optional Arabic letters) saved in an AES-256-CBC encrypted SQLite vault.

Electron React Vite SQLite

github.com/Ahshika/Password-Generator-V1

Wedding & Event Decor AI Bot

- Live webhook linking Messenger and WhatsApp Cloud API to Gemini; an Egyptian-Arabic chat that qualifies leads.

Node.js Express Gemini Meta API

github.com/Ahshika/wedding-decor-bot

Opti Nav — FiberSmart CAD Desktop

- Desktop CAD viewer that led to FiberLens: multi-document tabs, layouts, markup, parser fallback for DWG, DXF, STEP, IGES, STL.

Electron React TypeScript Zustand

github.com/Ahshika/Opti-Nav